

Portfolio Rebalancing Engine: Problem Analysis & Test Matrix

Concept Requirement:

Account ABC holds these securities - IBM, MSFT, ORCL, AAPL, HD

Account- ABC with \$100K in total asset (100% is vested)	Security	Target%	Current%	Target Variance	Unit price	Output - Number of shares to buy/sell
	IBM	20	10	-10	150	
	MSFT	20	20	0	90	
	ORCL	20	30	10	220	
	AAPL	20	20	0	450	
	HD	20	20	0	70	

Total Asset - \$100,000

target% indicates what % of total assets, you should hold in your acct.

current% shows what % of total assets, you currently have in your Acct.

target Variance shows how much deviation you have, (-ve) var means you need to buy (+ve) var means you need to sell.

Question: What do you have to do to get to zero target variance?

Answer: We must execute the exact calculated trade orders (BUY actions for negative variance and SELL actions for positive variance) across the securities to align the portfolio's current state with its target allocation.

Core logic:

Target Variance = Current % - Target %
Value Delta = Total Asset x (target variance /100)
Shares to trade = Value Delta / Unit Price

Portfolio Matrix:

Security	Target %	Current %	Target Variance	Value Delta (\$)	Unit Price (\$)	Calculated Shares	Action
IBM	20	10	-10	-10,000	150	-10000/150=66.6667	Buy 66.67 shares
MSFT	20	20	0	0	90	0/90=0	No Action (Hold)
ORCL	20	30	10	+10,000	220	10000/220=45.4545	Sell 45.45 shares
AAPL	20	20	0	0	450	0/450=0	No Action (Hold)
HD	20	20	0	0	70	0/70=0	No Action (Hold)

Manual Test Cases:

Functional Scenarios:

TC-01: Happy Path Rebalancing (Account ABC)

Setup: Total NAV: \$100K | IBM Var: -10% (Price: \$150) | ORCL Var: +10% (Price: \$220).
Expected Result: IBM: BUY 66.6667 shares | ORCL: SELL 45.4545 shares | All others: 0 shares.

TC-02: Zero Variance Baseline (Already Balanced)

Setup: Total NAV: \$100K | All asset variances = 0%.
Expected Result: All outputs generate 0 shares to trade (No actions taken).

TC-03: Boundary Condition (100% Structural Portfolio Shift)

Setup: Total NAV: \$100K | Security A (Current 100% to Target 0%) | Security B (Current 0% to Target 100%).
Expected Result: Security A: Sell all shares | Security B: Allocate full \$100K asset value to purchase maximum shares.

TC-04: High-Precision Matrix & Rounding Verification

Setup: Total NAV: \$100,000 | Target Variance: -10% (Value Delta: -\$10,000) | Unit Price: \$33.33.
Expected Result: System executes a BUY order of exactly 300.0300 shares (Validated to four decimal places).

Edge-Case Scenarios:

TC-05: Total Target Allocation Integrity Violation

Setup: Input an invalid target allocation matrix summing to 95% or 105%.
Expected Result: System rejects input with validation error: "Total target allocation must equal 100%"

TC-06: Compliance Breach (Zero or Negative Asset Valuation)

Setup: Set target security unit price to \$0.00 or -\$5.00.
Expected Result: System rejects input with validation error: "Unit price must be greater than zero"

TC-07: Asset Allocation Minimum Boundary (Zero Net Portfolio Value)

Setup: Total Portfolio Asset = \$0.00.
Expected Result: All trade orders yield 0 shares, no division-by-zero errors triggered.

Automated Test Execution Engine:

GitHub Repository Link: <https://github.com/Gippy-1514/portfolio-rebalancer>
CI/CD Pipeline Link: <https://github.com/Gippy-1514/portfolio-rebalancer/actions>

Extent Report Summary Metrics:

Matric Category	Statistics
Total Test Scenarios Executed	3 Scenarios testHappyPathAccountABC, testInvalidTargetAllocationSum, testInvalidUnitPrice
Execution Outcome Status	3 Passed / 0 Failed / 0 Skipped (100% Stability Rate)
System Environment	Java 23 Platform Runtime Container

